

Kanan Mahammadli

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EDUCATION

Middle East Technical University

Ankara, Turkey

Bachelor of Science in Mathematics (with Minor in Statistics), CGPA: 3.64

Sep. 2020 – June 2025

- **Ranking:** 5/139
- **Honors:** High Honor Student (4 semesters), Honor Student (3 semesters)
- **Coursework:** Calculus, Linear Algebra, Probability, Statistics, Optimization, Real/Complex Analysis, Machine Learning, Deep Learning, Reinforcement Learning, Sequence Models

PUBLICATIONS

- **Mahammadli, K., & Ertekin, S. (2024).** “Sequential Large Language Model-Based Hyper-parameter Optimization.” arXiv preprint arXiv:2410.20302. (Under Review at Knowledge-Based Systems Journal). Citations: 5.
 - Developed SLLMBO framework to enhance LLM-based hyperparameter optimization.
 - Designed hybrid LLM-TPE sampler outperforming LLM-only methods on 9/14 tasks.
 - Elevated Gemini model into top-tier optimizer using proposed hybrid strategy.

ACADEMIC RESEARCH EXPERIENCE

Undergraduate Researcher

July 2023 – June 2025

AI Lab, METU BILTIR Center (Prof. Seyda Ertekin)

Ankara, Turkey (Onsite)

- Developed SLLMBO framework for LLM-guided hyperparameter optimization.
- Worked on computer vision models (detection, segmentation).

Deep Learning Research Intern

Oct. 2022 – Nov. 2022

University College London (Prof. Yukun Hu)

London, UK (Remote)

- Worked on physics-informed neural networks project.

Computer Vision Research Intern

July 2022 – Sep. 2022

University of Houston (Prof. Ioannis Kakadiaris)

Houston, TX (Onsite)

- Worked on computer vision tasks involving object detection and representation learning, focusing on optimizing multi-scale models and similarity-based analysis.

RESEARCH PROJECTS

Early Sepsis Detection

Feb. 2024 – June 2024

- Supervised by Prof. Erdem Akagunduz
- 7-step-ahead forecasting for sepsis detection using sequential EHR data.
- Custom loss design to penalize late predictions and handle class imbalance.

Breast Cancer Classification via Data Augmentation

Mar. 2022 – July 2022

- Supervised by Prof. Seyda Ertekin
- Used Vision Transformer + class-specific augmentations to boost F1-score of minority classes.